Seth Ricks ECEN 470 PS03: 3.45(a)(b)(c), 3.46(a)(b)(Hint: see c) 3.53(a), 3.55

1/6

(3.45) Find TF below:

a.) Which meet an overshoot of $M_p \leq 5\%$?
b.) " " a rise time of $t_r \leq 0.5$sec?
c.) " " a setting time of $t_s \leq 2.5$sec?

$$G_1(s) = \frac{40}{s^2+4s+40}$$
$$G_2(s) = \frac{40}{(s+1)(s^2+4s+40)}$$
$$G_3(s) = \frac{120}{(s+3)(s^2+4s+40)}$$
$$G_4(s) = \frac{20(s+2)}{(s+1)(s^2+4s+40)}$$
$$G_5(s) = \frac{36040/401\,(s^2+s+401)}{(s^2+4s+40)(s^2+s+401)}$$

$$H(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$G_1(s)$: $4s = 2\zeta\omega_n s \rightarrow 2 = \zeta\omega_n$

$40 = \omega_n^2$

$\omega_n = \sqrt{40} = 2\sqrt{10}$

$2 = \zeta(2\sqrt{10})$

$\zeta = \frac{1}{\sqrt{10}}$

$t_r = \frac{1.8}{\omega_n} = \frac{1.8}{2\sqrt{10}} = 0.285$ ✓

$t_s = \frac{4.6}{\zeta\omega_n} = \frac{4.6}{(\frac{1}{\sqrt{10}})(2\sqrt{10})} = 2.3$ ✓

$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$

$= e^{-\pi/\sqrt{10}/\sqrt{1-(1/\sqrt{10})^2}}$

$= 0.35 \rightarrow 35\%$ X

$M_p \leq 5\%$? NO
$t_r \leq 0.5$ sec? Yes
$t_s \leq 2.5$ sec? Yes

2/6

$G_2(s) = \frac{40}{(s+1)(s^2+4s+40)}$ Poles: $s = -2 \pm j6$, $s = -1$

* within factor of 4 → significant effect *

For $G_2(s)$, since pole is within factor of 4 of the real part of the original poles, overshoot will decrease, rise time will increase, and settling time will increase.

$M_p \downarrow \leq 5\%$? → probably ✓
$t_r \uparrow \leq 0.5$ sec? → probably not ×
$t_s \uparrow \leq 2.5$ sec? → probably not ×

$G_3(s) = \frac{120}{(s+3)(s^2+4s+40)}$

Original: $6 = -2$
new: $s = -3$
$Re\{p\} < 46$: yes
-3 further left than -2 → faster
$M_p \downarrow \leq 5\%$? → probably ✓
$t_r \uparrow \leq 0.5$ sec? → probably not ×
$t_s \uparrow \leq 2.5$ sec? → probably not ×

$G_4(s) = \frac{20(s+2)}{(s+1)(s^2+4s+40)}$

Original: $6 = -2$ closer to origin
new p: $s = -1$
new z: $s = -2$
} → within factor of 4 = yes
$M_p \downarrow$ → ✓
$t_r \uparrow$ → ×
$t_s \uparrow$ → ×

$G_5(s) = \frac{36040/401\,(s^2+s+4011)}{(s^2+4s+40)(s^2+s+901)}$ Original: $6 = -2$
New poles: $s = -0.5 \pm j30$
New zeros: $z = -0.5 \pm j20$

both -0.5, balance each other out.
$M_p$ ×
$t_r$ ✓ $t_s$ ×

After MATLAB:

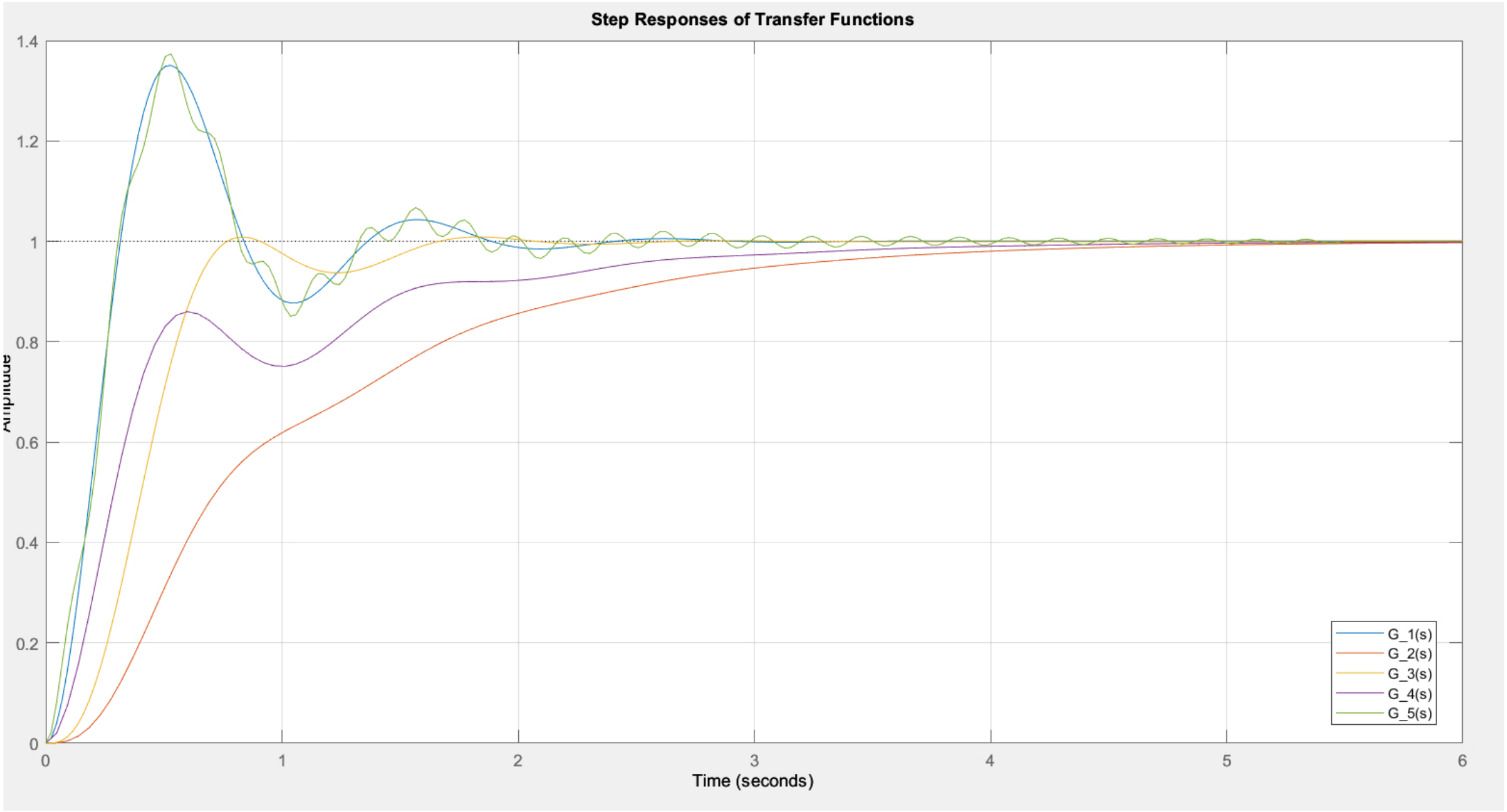
3/6

| | $M_p \leq 5\%$ | $t_r \leq 0.5$ sec | $t_s \leq 2.5$ sec |
|---|---|---|---|
| $G_1$ | × | ✓ | ✓ |
| $G_2$ | ✓ | × | × |
| $G_3$ | ✓ | ✓ | ✓ |
| $G_4$ | ✓ | × | × |
| $G_5$ | × | ✓ | ✓ |
| | ↑ a.) | ↑ b.) | ↑ c.) |

```
% Define the Laplace variable s
s = tf('s');

% Define the five transfer functions
G1 = 40 / (s^2 + 4*s + 40);
G2 = 40 / ((s + 1) * (s^2 + 4*s + 40));
G3 = 120 / ((s + 3) * (s^2 + 4*s + 40));
G4 = (20 * (s + 2)) / ((s + 1) * (s^2 + 4*s + 40));
G5 = ((36040/401) * (s^2 + s + 401)) / ((s^2 + 4*s + 40) * (s^2 + s + 901));

% Plot all step responses together for comparison
figure;
step(G1, G2, G3, G4, G5);
legend('G_1(s)', 'G_2(s)', 'G_3(s)', 'G_4(s)', 'G_5(s)', 'Location', 'best');
grid on;
title('Step Responses of Transfer Functions');
```


Step Responses of Transfer Functions
Amplitude
Time (seconds)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
0
1
2
3
4
5
6
G_1(s)
G_2(s)
G_3(s)
G_4(s)
G_5(s)

3.46 Consider following two nonminimum phase systems.

$$G_1(s) = -\frac{2(s-1)}{(s+1)(s+2)} \qquad G_2(s) = \frac{3(s-1)(s-2)}{(s+1)(s+2)(s+3)}$$

a.) Sketch unit step response for $G_1(s)$ & $G_2(s)$, paying close attention to the transient part of response
b.) Explain difference in behavior of two responses as it relates to the zeros

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* step: $\mathcal{L}\{u(t)\} = \frac{1}{s}$

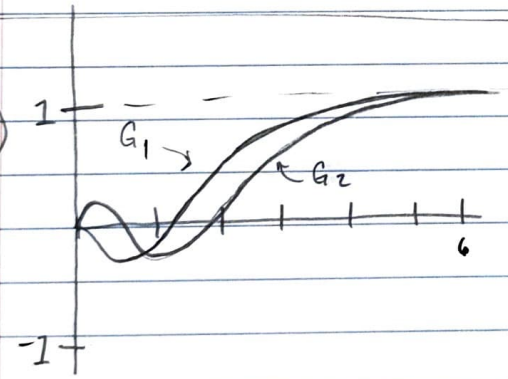
$$G_1(s)\cdot\frac{1}{s} = \frac{-2(s-1)}{s(s+1)(s+2)} = \frac{1}{s} - \frac{4}{s+1} + \frac{3}{s+2}$$

$$g_1(t) = [1 - 4e^{-t} + 3e^{-2t}]u(t)$$

$$G_2(s)\cdot\frac{1}{s} = \frac{3(s-1)(s-2)}{s(s+1)(s+2)(s+3)} = \frac{1}{s} - \frac{9}{s+1} + \frac{18}{s+2} - \frac{10}{s+3}$$

$$g_2(t) = [1 - 9e^{-t} + 18e^{-2t} - 10e^{-3t}]u(t)$$

a.)




b.) $G_1$ = odd # of zeros = undershoot
$G_2$ = even # of zeros = overshoot

3.53 Use Routh's stability criterion to determine how many roots w/ + real parts.

a.) $s^4 + 8s^3 + 32s^2 + 80s + 100 = 0$

Routh array:

$s^4$: 1 32 100

$s^3$: 8 80

$s^2$: $\frac{(8)(32)-(1)(80)}{8} = 22$ $\frac{8(100)-(1)(0)}{8} = 100$

$s^1$: $\frac{(22)(80)-8(100)}{22} = 43.63$

$s^0$: 100

No poles in RHP

0 roots with pos real parts

3.55 TF: $KG(s) = \frac{K(s+4)}{s[(s+0.5)(s+1)(s^2+0.4s+4)]}$ 6/6

Using Routh's stability criterion, determine range of K for which system is stable when $1+KG=0$.

Char eq: $1+KG(s)=0$

$$1+\frac{K(s+4)}{s[(s+0.5)(s+1)(s^2+0.4s+4)]}=0$$

$$s[(s+0.5)(s+1)(s^2+0.4s+4)]+K(s+4)=0$$

$$s^5+1.9s^4+5.1s^3+6.2s^2+2s+Ks+4K=0$$

$$s^5+1.9s^4+5.1s^3+6.2s^2+(2+K)s+4K=0$$

Routh array:

| | | | | |
|---|---|---|---|---|
| $s^5$: | 1 | 5.1 | $(2+K)$ | ← (odd powers) |
| $s^4$: | 1.9 | 6.2 | $4K$ | ← (even powers) |
| $s^3$: | 1.837 | $2-1.105K$ | | |
| $s^2$: | $4.13+1.14K$ | $4K$ | | |
| $s^1$: | $\frac{-1.26K^2-9.63K+8.26}{4.13+1.14K}$ | | | |
| $s^0$: | $4K$ | | | |

$\frac{1.9(2+K)-4K}{1.9} = 2-1.105K$

$\frac{(1.837)(6.2)-1.9(2-1.105K)}{1.837}$

$\frac{11.389-3.8+2.099K}{1.837}$

$4.131+1.143K$

$\frac{1.837(4K)-1.9(0)}{1.837}=4K$

$\frac{(4.13+1.14K)(2-1.105K)+(1.837)(4K)}{4.13+1.14K}$

$\frac{-1.26K^2-9.63K+8.26}{4.13+1.14K}$

every element in 1st column must be >0

$s^0$: $K>0$
$s^1$: $K<0.78$
$s^2$: $K>-3.62$
$s^3$: $K<1.81$

$\boxed{0<K<0.78}$